

Response plan for ja-2026-12808m

Internal note (not the letter to Coley)

19 August 2026

Coley rejected the manuscript; resubmission is invited by 13 February 2027. Reviewer 1 asked for minor revision and put novelty in the top 5%. Reviewer 3 asked for major revision and put significance in the top 5%. Reviewer 2 said publish elsewhere, on impact, not on correctness.

We will not run every requested calculation. From the co-author note we keep five jobs: a constructed out-of-distribution test, one hydrogen-bonded system, one comparison with a modern ML potential, one density above PBE, and a rewrite of the Abstract, the last paragraph of the Introduction, and the first paragraph of the Discussion so the paper argues that $n(\mathbf{r})$ is available at every MD step.

This note walks every numbered comment in reviewer order. The italic sentence is their point. The bullets are what we will do, or have already done. The table at the end is the work list.

1 Reviewer 1

R1.1 Hydrogen-bonded size extrapolation

The systems are still simple; how does this work when hydrogen bonds matter?

- Train on water clusters $n = 2-6$, test on $n = 8, 10, 12$ (the protocol they named).
- Label the same geometries at PBE-D4 and ω B97M-V so the H-bond figure and the hybrid table share a dataset. Labelling is running: PBE train is done, ω B97M-V train is in flight.
- Skip NMA, methanol-water, and AcAc. Malonaldehyde covers the intramolecular proton-transfer motif they also listed.

R1.2 Out-of-distribution conformations

OOD conformations are missing, and a second MD trajectory is not proof of non-overlap.

- Built a ladder and measured it: in-distribution test (control), size hold-out (weak, and we say so), ice cutouts, low-density droplets, malonaldehyde proton-transfer scan.
- Separation: Bhattacharyya overlap plus nearest-neighbour distance in CV space, in units of the training set's own spacing. The figure is density error versus that distance.
- Ethanol high- T / strain OOD stays as a cheap extra, not the panel we lead with.

R1.3 Polythiophene length and Figure 6

The polythiophene discussion is too long, and Figure 6 carries little.

- Shorten the section. Move most of Figure 6 to the SI. Keep one oligomer IR comparison in the main text. Manuscript only.

R1.4 Figure 7

Figure 7 is standard and belongs in the SI.

- Move it. Manuscript only.

R1.5 Dataset documentation and train/test overlap

SI S2.2 is thin, and “a separate MD run” does not prove the test set is non-overlapping.

- Print the RMSD numbers we have: resorcinol test frames sit inside the training basin (median min-RMSD 0.093 Å, all 1000 below 0.2 Å); ethanethiol and thiophene tests do not.
- If we find the force field, SHAKE, and trajectory length, we add them; if not, we say they were not recovered. The new water splits already have the same overlap treatment.

R1.6 Duplicated timing

The 755 ms versus 5 min timing appears twice.

- Keep one instance. Ethanol CPU: 1.62 s/step. GPU: 467 ms/step.

R1.7 Unsupported dielectric claim

The dielectric / hydrogen-bond-network sentence on p. 25 is not supported.

- Delete it. If the water-cluster results later support a narrower claim about H-bonds, that claim goes next to those results, not in the Discussion as a promise.

2 Reviewer 2

R2.1 Hazra–Sanvito density-to-IR workflow

Hazra, Patil, and Sanvito already go density to energy, forces, and IR.

- Cite *J. Chem. Phys.* 2025, 163, 174104.
- Their density is a Jacobi–Legendre cluster expansion on uracil. Ours is an equivariant network with density fitting and a SAD correction, plus size, OOD, and H-bond tests they did not. Citation only.

R2.2 Faster general IR methods

MACE4IR, AIMNet-2, TranSpec, and AIQM are faster and work across chemical space.

- Compare transferable MLIPs and xTB on the molecules we already trained (expand Figure 2).
- We will not retrain on Crambin, paracetamol, or FGG. TranSpec maps SMILES to a spectrum; that is a different task and we say so.

R2.3 Oligomer size extrapolation

Training through the hexamer and testing longer oligomers is a weak extrapolation, and the error grows.

- Agree that this test is weak.
- Harder tests: ice, droplets, malonaldehyde scan. Size-OD overlap is about 0.20, above the 0.05 bar we set for ice and droplets, and we report that.

R2.4 Companion network and energy conservation

How does the companion network work, and is the AIMD energy conserving?

- Figure 1: $\{\mathbf{R}_I\} \rightarrow \text{DenSNet} \rightarrow n(\mathbf{r}) \rightarrow \text{EnergyNet} \rightarrow E$, with $F_I = -\partial E / \partial \mathbf{R}_I$. Gradients pass through the density features.
- Quote NVE drift only after the energy head matches the 2024 checkpoint (96w7KyGG). Current 0.18–0.30 eV/ps numbers are from a mismatched head and stay out of the paper.

R2.5 Who produces dipoles

Page 4 implies the companion network produces electronic observables.

- Dipoles come from the density (`int1e_r`). The companion head outputs energy and forces. Fix page 4.

R2.6 One framework or two networks

The text says one framework; there are two networks.

- Both modules run at every MD step. IR does not need a post-processed trajectory if the dipole integrator is on. Drop “single network”.

R2.7 Electron-count normalization versus delta-learning

If normalizing by electron count already weighted core and valence, why delta-learning?

- Normalization is a loss weight so raw grid errors in the core do not dominate. SAD changes the target to $\rho - \rho_{\text{SAD}}$. They are not substitutes. Keep both.

R2.8 Softplus and signed Δn

Delta-density is negative in places; how does that sit with a softplus?

- Softplus is applied so the reconstructed total density stays non-negative. Δn itself is signed.
- About half of the DF coefficients on water and ethanethiol tests are already negative (46.7% and 51.0%).

R2.9 AIMD length

State the AIMD length in the main text.

- Ethanol AIMD on disk is 10 ps. Put every trajectory length in the main text (same job as R3.6).

R2.10 Density-error units

Density errors have no units.

- MAE in e/a_0^3 and relative MAE $\int |\Delta\rho| dV/N_e$.
- We will not invent a density analogue of 1 kcal/mol. Tie density error to dipole error and then to IR intensity.

R2.11 Equilibrium structures

The manuscript claims DFT-like equilibria without numbers.

- After the energy head is restored, report geometry RMSD versus DFT on a few small molecules.
- Do not quote the current ethanol energy of ~ 0.24 eV; that is an architecture mismatch.

R2.12 Missing density ML citation

Cuevas-Zuñiría and Pacios, J. Chem. Inf. Model. 2020, is missing.

- Add it to the introduction.

R2.13 Corresponding authors

Corresponding authors disagree between the manuscript and the SI.

- Make them match. Confirm the list with Tuckerman, Burke, and Müller before resubmission.

3 Reviewer 3

R3.1 Hybrid DFT densities

All reference densities are PBE; try PBE0+MBD or ω B97M-V/def2-TZVPD.

- Nothing in the architecture is PBE-specific.
- Label the same water and malonaldehyde geometries at ω B97M-V/def2-TZVPD and at PBE-D4; one table. No hybrid MD, no hybrid IR, no retraining of ethanol or thiophene at hybrid.

R3.2 Comparisons with ML force fields

Compare SO3LR, MACE-OFF, GFN-xTB, DFTB, and the Crambin / paracetamol / FGG IR suites.

- Expand Figure 2 on our molecules.
- Decline Crambin, paracetamol, and FGG: those papers test transferable MLIPs, and DenSNet is fitted per system.

R3.3 Interpreting density errors

A density MAE of order 10^{-3} is hard to interpret; add relative errors and uncertainty.

- Relative MAE, plus density error against dipole error and IR intensity error.
- A three-seed ensemble on water if it is cheap. Uncertainty quantification is not a separate project.

R3.4 Cost and magnitude of delta-learning

Delta-learning needs a cost and a magnitude, not just a lower MAE.

- Time the SAD lookup. Histogram $|\Delta\rho|/\rho$.
- Train direct and delta models on water and compare them with relative errors (configs already exist).

R3.5 Hyperparameters and cutoff

Architecture parameters look un-optimized, and cutoff is known to matter.

- Write down the paper cutoffs (5.0 Å thiophene, 15 Bohr small molecules).
- Sweep cutoff {4, 5, 6, 8} Å on water at PBE only. No grid search.

R3.6 IR convergence with trajectory length

“Sufficient duration” is not a number.

- State the lengths. Plot IR versus 50, 100, 200, 500 ps on ethanol and thiophene 2-mer (those jobs exist). Report peak position and integrated area as a function of length.

R3.7 General-purpose density model

Could DenSNet become a general-purpose density model, perhaps with active learning?

- Discussion only. Present models are system-specific. A later path is large density datasets, pretraining, fine-tuning, and active learning.
- We will not build that model in this revision. OMol/CSH can be mentioned as work already running, not as a result of the paper.

R3.8 Spectral language

Peak positions agree better than intensities; stop saying the spectra match across the full range.

- Peak positions in named windows. Intensities and areas agree less evenly. The recovered T_{EX} already has this wording; keep it.

R3.m1 Two-stage training cost

Training cost for the two-stage procedure, especially on the larger systems.

- GPU-hours in the SI.

R3.m2 Architecture in Figure 1

Figure 1 should include the architecture, not only the workflow.

- Same figure as R2.4.

R3.m3 Other density-derived properties

Which other electronic properties could come from the predicted density?

- Hirshfeld charges, ESP, polarizability: list as future observables. No new calculations.

4 To-do

Columns: ID, task, kind, status, depends on.

Table 1: Work list for the resubmission.

ID	Task	Kind	Status	Depends on
<i>Compute</i>				
R1.1	Water PBE labels, train, ID and size-OOD metrics	DFT	running	–
R3.1	Water ω B97M-V labels, same-geometry hybrid table	DFT	running	R1.1 geometries
R1.2, R2.3	Ice, droplet, malonaldehyde labels, then error versus distance	DFT	running	train checkpoint
(ops)	Watchdog running so training actually starts	ops	down	–
R3.5	Cutoff sweep at PBE	analysis	waiting	water PBE train
R2.4, R2.11	Restore energy head, then NVE and geo-opt RMSD	MD	blocked	2024 checkpoint
R3.6	Ethanol and thiophene IR versus length	MD	jobs exist	trajectories
R2.2, R3.2	Figure 2 GPU timings versus MACE-OFF, AIMNet2, xTB	MD	partial	GPU QOS
R2.10, R3.3, R3.4	Relative density error, $\Delta\rho$ histograms, SAD timing	analysis	scripts ready	trained water model
<i>Manuscript (can start now)</i>				
–	Abstract, last Intro paragraph, first Discussion paragraph: $n(\mathbf{r})$ at every MD step	manuscript	not started	–
R1.3, R1.4	Shorten polythiophene; move Figures 6 and 7	manuscript	outlined	–
R1.7, R3.8	Delete dielectric sentence; keep softer spectral wording	manuscript	TeX already edited	–
R2.4–R2.6, R3.m2	Architecture in Figure 1; dipoles versus energy head; two modules	manuscript	outlined	–
R1.5, R2.9, R2.10	SI S2.2, overlap table, AIMD lengths, units	manuscript	overlap computed	–

ID	Task	Kind	Status	Depends on
R2.1, R2.12, R2.2	Cite Hazra–Sanvito, Cuevas-Zuiviría, Pracht, TranSpec, AIQM	manuscript	not started	–
R2.13, R3.m1	Corresponding authors; two-stage GPU-hours in SI	manuscript	not started	names from PIs
<i>Decline</i>				
R3.2	Crambin, paracetamol, FGG	decline	–	–
R3.1	Retrain ethanol, thiophene, or polythiophene at hybrid	decline	–	–
R3.7	Universal DenSNet or active-learning campaign as a paper result	decline	–	–
R3.5	Full hyperparameter search	decline	–	–
R1.3	New polythiophene experiment	decline	–	–